

Technical Appendix: The Harness, Not the Hierarchy

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A. Protocols by Stage

All continuous stages share the substrate: a 2-D point robot, worlds with procedurally generated obstacles, a probabilistic roadmap (PRM) of ~ 192 nodes with $k=7$ connectivity, and A^* evaluated at a per-suite *binding budget* of node expansions calibrated so the algorithmic baseline is neither saturated nor hopeless. Unless noted, training labels are exact graph-Dijkstra costs on the PRM (static) or exact backward space-time Dijkstra values over (node, t) (dynamic). Headline continuous runs are local single-GPU (RTX 5090) validations with one primary training seed unless stated; the discrete program ran on a remote fleet.

B. Full Static and Dynamic Suite Tables

Tables 2 and 3 give the complete per-suite C7 static and C8 dynamic results summarized in the main paper.

C. Transfer and Composition Details

Representative C9 HRM curves (ratio at success): maze-dense LoRA 0.650 (1.00) at $K=1$ to 0.730 (0.97) at $K=16$; maze-dense full FT 0.744 (0.76) to 0.571 (0.99) at $K=16$ and 0.552 at $K=32$; scratch 1.008 (0.37) to 0.808 (0.80); rooms-large full FT 1.128 (0.37) at $K=1$ to 0.500 (0.92) at $K=8$; bugtrap LoRA degrades from ≈ 0.696 (0.83) to 0.950 (0.48) at $K=32$ —LoRA is usually but not universally base-preserving. C9H (matched compute): bounded-minus-unbounded LoRA median ratio delta 0.000 ± 0.008 across 27 cells (22/27 exactly zero; max $|\Delta|$ 0.028); rooms-large field U-Net full FT reaches 0.404 at 0.97 success from 0.982 zero-shot while conv-LoRA stays at 0.992–1.002. C9B (dynamics): maze-dense aware U-Net zero-shot 0.145 (0.60), LoRA $K=16$ 0.059 (0.97), full FT $K=1$ 0.112 (0.70); aware-minus-blind success deltas at full-FT $K=16$ are 0.00 or -0.05 in all nine cells. C10: RBF-merge minus zero-shot expansion-ratio deltas by target/backbone: -0.022 , -0.012 , -0.024 , $+0.082$, $+0.012$, $+0.116$; weight-space vs. prediction-space deltas lie within $[-0.012, +0.007]$; own-axis RBF mass 0.986–0.998.

D. C11 Gates, Collapse Forensics, and Scale

Collapse forensics. Five of 33 recurrent/ACT arm-cells show constant-prediction or partial seed collapse (HRM-trace A8/C8, ON-LSTM B0, HRM-v2 B0, two ON-LSTM

A8 seeds), with a reproducible loss signature at the normalization cap. The largest recurrent deficits (HRM-trace ratios 1.526 at A8, 1.600 at C8) are collapse measurements, not expressiveness estimates. Removing $K=0$ padding *worsens* recurrent predictions: models used pad steps as extra recurrent compute.

Scaled addendum. A 12-run `unet_film_big` vs. `hrm_trace_big` comparison (complete; 2,700 eval rows) preserves the ordering: completion 0.804 vs. 0.736 at $K=2$ and 0.413 vs. 0.307 at $K=8$; mean state MAE 0.395 vs. 0.447 ($K=2$) and 0.906 vs. 1.855 ($K=8$). At $K=8$ the scaled HRM exactly matches the leg-sum baseline’s completion and mean expansions (0.307; 701.34)—persistent collapse, not recovered hierarchy.

E. C12 Detail

C12-A (memory headroom, then learned pilot): constructed decision-relevant aliasing at 71.3% of paired decisions; privileged mode/history diagnostic $+0.467$ completion and -65.1% collision-adjusted regret (world-bootstrap 95% CI 63.0–66.9%); oracle completion 0.975. The one-seed matched temporal-hierarchy pilot fails G1 (forecast), G2 (planning), and G3 (persistent carry); overall *strong negative*. C12-B (tied refiner): cycle-1 $\rightarrow$ 8 normalized burden improves monotonically in every cell (A/K^2 0.859 $\rightarrow$ 0.815; A/K^8 0.624 $\rightarrow$ 0.576; C/K^2 0.867 $\rightarrow$ 0.777; C/K^8 0.576 $\rightarrow$ 0.517; all cycle-1/8 world-bootstrap intervals separate), but the K^8-K^2 gain is $+0.0038$ (CI $[-0.0199, 0.0281]$) on A and -0.0305 ($[-0.0726, 0.0096]$) on C—no dose-response. Tied beats shallow and untied on C/K^8 (deltas 0.0177/0.0076, BH $q=6.7 \times 10^{-5}$) but loses to untied on A/K^8 ($q=0.0168$). Bellman residual *worsens* over cycles in every cell while burden improves; solved cycle-8 paths average 1.4–2.2% above oracle cost. $K=16$ was dropped before training by a construction gate (only 2/300 and 1/300 valid missions).

F. Discrete Program Detail

Forecasting. Matched 100-episode protocol: HRM-3M/10M 71/100 vs. LSTM 67–69/100 (no uncertainty reported; two-episode matched gap). Structured Preset M+ (four 100-episode suites): ON-LSTM-10M 0.3875 mean success vs. HRM-10M 0.2650 and HRM-3M 0.200; ON-

LSTM-3M also has the lowest rollout MSE (0.1257/1.1269 one-/five-step vs. 0.3973/2.6438 for HRM-3M).

Additive transfer (clean v3, 22 suites $\times$ 3 budgets). Manhattan 0.590 success / 126,707 mean expansions; HRM full-FT 0.585 / 128,013; ON-LSTM full-FT 0.564 / 127,228; HRM LoRA 0.554 / 140,852; ON-LSTM LoRA 0.514 / 145,282. Every learned arm $\leq$ matched baseline.

Pooled multitask. HRM pooled base 0.612 success (+2.11pp) / 122,184 expansions vs. Manhattan 0.591 / 126,630—the strongest completed discrete additive positive; only 1 of several specialists beats its matched four-suite baseline (+2.0pp; others -0.42 to -20.25 pp).

Focal ranking (local pilot, 3–8 seeds). Magnitude is scale-flat despite near-perfect ordering (correlation 0.987–0.994; predicted/true mean 0.73/0.42/0.27/0.19 at 64/128/192/256). At $w=1.0$: HRM A128-static ratio 0.85 with success 0.62 $\rightarrow$ 0.75; A192 0.94 (0.75 $\rightarrow$ 0.75); A128-dynamic 0.93 (0.62 $\rightarrow$ 0.62); ON-LSTM A128/A192 0.85 (0.75 $\rightarrow$ 0.75). At $w=1.05$, A128-dynamic improves further but A192 success regresses 0.75 $\rightarrow$ 0.62; $w=1.0$ is the evidence-safe operating point. Experts exactly match pooled bases in tested cells.

G. Validity Forensics

The frozen-halt artifact. A 27.27M-parameter direct maze solver reported 96.64% token accuracy, 25.40% exact accuracy, and “74.60% halt accuracy” with 16 average adaptive-computation steps. Audit: 74.60 = 100 – 25.40 exactly—the halt logit was frozen negative (a constant-false classifier) because the training loop omitted the halt loss and back-propagated only the final segment. After repairing the loss path (verified live: halt loss 0.14 $\rightarrow$ 0.06, token accuracy 90–95%), exact accuracy remained 0.000 at the chosen stop point; the repaired run demonstrates mechanism liveness, not task convergence. Bit-exact forward parity with the reference implementation was separately established. We recommend the 100– x pattern as a standard red flag for adaptive-computation metrics.

The cap-collapse signature. C5’s scalar HRM collapse sits at a loss value pinned by the residual cap (2.612 in that setup, with constant predictions at the cap). The same class of signature reappears in C11: collapsed cells share an *identical* final loss (1.1397, exactly the constant-cap loss under that normalization), enabling automatic collapse detection instead of misreading collapsed cells as capacity evidence.

Pad steps as compute. In C11, removing $K=0$ padding tokens *degraded* recurrent predictions—models had co-opted padding as additional recurrent iterations. Sequence-model comparisons that alter padding alter compute.

H. Statistics

Success is compared with paired McNemar tests, BH-corrected within stage families. Expansion effort is compared only on matched solved worlds via median ratios with bootstrap percentile CIs (10k–20k resamples); from C12 onward, bootstraps are world-clustered by design. C9–C11 originally pooled repeated evaluations of fixed test worlds

across adaptation/model seeds; Section I reports the world-clustered reanalysis of every quantity the main paper quotes from those stages. Matched-solved ratios condition on joint success and can favor arms that solve only easier worlds; we therefore always report success and ratios separately and never combine them into a single score. Negative results are failures to observe an effect under the tested protocol, not equivalence claims; no equivalence tests were run.

I. World-Clustered Reanalysis

Units are TEST worlds; adaptation/model seeds are averaged within worlds before inference; success deltas use 10,000-resample world bootstraps; expansion ratios are medians of per-world ratios over matched worlds at the binding budget in additive mode; the C11 halting test aggregates model seeds within worlds and permutes mission length within config (worlds are independent across K cells; 20,000 stratified permutations). Estimator conventions differ slightly from the published record-level curves, so point values differ modestly; the question is verdict-level agreement. A methodological caution surfaced during this work: a first pass that accidentally pooled additive and focal evaluation modes attenuated the apparent C9 transfer effects by roughly half—mode/aggregation mixing alone can manufacture or destroy an effect of the size under study.

C11 halting. Record level reproduces the published value ($\rho = -0.4066$, $n=675$); world level strengthens it ($\rho = -0.5778$, $n=225$, stratified permutation $p=5\times 10^{-5}$). World-level means 6.78/7.01/5.30 at $K=2/4/8$.

C11 vs-MLP ratios. World-clustered ratio-of-means CIs exclude parity at $K \in \{0, 2\}$ in config A for both U-Net (0.734/0.846) and GNN (0.822/0.923), and in config B at $K=0$ (0.904/0.818) plus B/ $K=2$ for the GNN (0.805); config C never separates; every $K \in \{4, 8\}$ cell includes parity.

C9H bound-vs-capacity. 27 cells: median bounded-minus-unbounded world-level ratio delta +0.0000; 15/27 exactly zero; sd 0.0077; max $|\Delta|$ 0.0370—the capacity attribution stands.

C9B aware—blind (full FT, $K=16$). Eight of nine point deltas are zero or negative; the single positive (+0.017 [0.000, +0.050]) does not exclude zero; no cell shows a significant aware advantage.

C10 composition. World-paired deltas vs. zero-shot: four cells ≈ 0 with CIs crossing zero; the two ON-LSTM rooms cells are significantly *worse* (+0.089 [+0.061, +0.121] and +0.098 [+0.076, +0.123]), matching the published point deltas and upgrading the null to “never better, sometimes significantly worse.”

No preregistered verdict flips under any of these recomputations.

J. Reproducibility

From C7 onward every stage freezes a written design (gates, budgets, analysis grain) before execution; artifacts (check-points, feature caches, cohort manifests, raw rows, verdicts, reports) are hash-bound in integrity manifests; confirmation cohorts are generated only after freezing model, integration,

and evaluation code; failed development gates block confirmation rather than triggering retuning. Duplicate-evaluation determinism is enforced where preregistered (byte-identical traces; identical decision rows). The final-stage follow-up on persistent planning-time state is preregistered with a fully test-verified harness (158 focused-plus-neighboring tests; hardware preflight). Its first staged execution ended at a preregistered execution-integrity hard stop—the harness’s independent reanalysis disagreed with the primary comparison computation, voiding all mechanism fields—and the named-defect repair and fresh-fingerprint rerun were in progress at submission time. It contributes no outcome evidence and is cited only as future work; we note the stop itself as the integrity machinery functioning as designed. Code, preregistrations, per-stage raw tables, and integrity manifests will be released; an anonymized archive accompanies the submission.

Stage	Protocol essentials
C1–C4	9 easy suites, $n=80/40$ worlds, budgets 100–200; pooled bases, TaskLoRA experts, RBF mixtures.
C5	calibrated hard maze/dense/rooms; 160 train / ~ 80 eval worlds; budgets 128–168; scalar residual target with cap.
C6	goal-conditioned raster residual field (8 input channels), sampled at PRM nodes; 96/40 worlds; budgets 128–168.
C7	6 suites (3 held-out families); 96/24 worlds per suite; binding budgets 140/24/56; scalar+field $\times$ additive+focal matrix; exact-Dijkstra oracle.
C8	deterministic moving circular obstacles; space–time PRM; aware ($W=8$ future window) vs. blind ($W=0$) twins; 53 train / 20 eval worlds/suite; 1,129,536 scalar samples.
C9/C9H	adapt C7 pooled bases to 3 held-out families; $K \in \{0..32\}$ labeled worlds; LoRA(r8, bounded/unbounded), full FT, scratch; C9H matches compute (10 epochs, $LR 2 \times 10^{-4}$); 30 fixed test worlds/target.
C9B	same protocol on C8 dynamic sources (aware+blind), 486 adapters, 3 targets, $K \in \{1, 4, 16\}$.
C10	16 rank-8 experts on 2 family axes; zero-label transfer to 3 interior targets; RBF/nearest/uniform weight-merge and prediction-mix composition.
C11	compositional missions (waypoint sequences; key/door), configs A/B/C, mission length $K \in \{0, 2, 4, 8\}$; 40 train / 25 test worlds per cell; 6 learned arms $\times$ 3 seeds (198 checkpoints; 54,450 eval rows); exact mission oracle and leg-sum baseline; forced-compute and would-halt probes.
C12	(A) hidden slow/fast regime dynamics, memory-headroom gate, matched temporal-hierarchy pilot; (B) tied vs. untied product-graph refiner, cycles 1–8, configs A/C $\times$ $K \in \{2, 8\}$, 3 seeds, 48 checkpoints, two 3,200-row test tables.
C13	current-state heuristic under bounded observation (radius 0.20 local subgraph); behavior-derived fresh-start rollout labels (median of 3); staged falsification ladder (A–L) then frozen 144-world confirmation (M); architecture substitutions (N/O).
Discrete	20×20 to 256×256 grids; Manhattan-baseline A^* ; forecasting (dynamic obstacles), additive residual transfer (22 suites $\times$ 3 budgets $\times$ 100 episodes), pooled multi-task bases, learned focal ranking at $w \in \{1.0, 1.05, 1.1\}$.

Table 1: Protocol essentials by stage.

Suite	Euclid succ.	Field-HRM succ.	Ratio [95% CI]
Maze	0.583	1.000	0.521 [0.450, 0.627]
Maze-dense	0.250	0.958	0.804 [0.707, 0.829]
Rooms	0.375	0.958	0.829 [0.775, 0.885]
Spiral	0.250	0.917	0.850 [0.742, 0.919]
Bugtrap	0.458	0.750	0.714 [0.533, 0.800]
Rooms-large	0.417	0.750	0.839 [0.646, 1.222]

Table 2: C7 static suites at binding budgets: success and matched-solved median expansion ratio (field HRM vs. Euclid). Success gains are BH-significant in 4/6 suites (maze, dense, rooms, spiral); bugtrap and rooms-large reach corrected significance only for other learned arms. Scalar-HRM ratios span 0.427–0.822. Matched n ranges 6–14.

Suite (arm)	Succ. E→L	Ratio [95% CI]	Matched n
Crossing (U-Net)	0.30→1.00	0.258 [0.132, 0.769]	6
Maze (U-Net)	0.30→1.00	0.064 [0.046, 0.088]	6
Maze-dense (U-Net)	0.05→0.75	0.278 (no CI)	1
Rooms (U-Net)	0.30→1.00	0.096 [0.038, 0.191]	6
Rooms-large (U-Net)	0.75→0.90	0.380 [0.132, 0.591]	14
Spiral (field HRM)	0.20→0.90	0.046 [0.038, 0.073]	4

Table 3: C8 heavy dynamic run, selected strongest arms: Euclid→learned success and matched-solved ratios. Corrected success evidence holds on five suites; rooms-large is supported mainly by expansion evidence. Dense-maze has matched $n=1$ and supports no expansion inference.

Gate	Verdict
K=0 continuity	FAIL, diagnosed as false premise: C7 had no MLP arm and did not give every architecture the C11 input; real shallow separation plus collapsed recurrent cells violate the all-tie assumption. Measurement layer verified.
G1 dose-response	Negative: no structured arm beats MLP at ≥ 2 mission depths with non-decreasing gap.
G2a forced compute	Negative: HRM-v2 $k=1/2/4/8$ expansion and MAE curves flat in every $K=8$ config.
G2b learned halting	Negative, inverted: pooled Spearman $\rho = -0.407$, permutation $p \approx 5 \times 10^{-4}$, $n=675$; mean would-halt steps 6.79/7.01/5.30 at $K=2/4/8$.
G3 closure	Neither preregistered branch: global-input arms separate at shallow K (U-Net/GNN vs. MLP ratios ≈ 0.67 – 0.87 ; state MAE 0.151/0.171 vs. 0.251 at $A/K=0$), but no arm converts depth into growing advantage.

Table 4: C11 preregistered gates.

Cell (HRM)	Δ success [CI]	world ratio [CI]
dense LoRA $K=1$	+0.467 [+0.300, +0.633]	0.686 [0.590, 0.774]
dense LoRA $K=16$	+0.433 [+0.267, +0.600]	0.786 [0.694, 0.891]
dense full $K=1$	+0.227 [+0.080, +0.380]	0.789 [0.751, 0.836]
dense full $K=16$	+0.460 [+0.287, +0.633]	0.610 [0.549, 0.670]
dense scratch $K=1$	−0.167 [−0.300, −0.047]	1.035 [1.029, 1.039]
rl. full $K=1$	−0.167 [−0.340, +0.013]	1.293 [1.170, 1.503]
rl. full $K=8$	+0.387 [+0.227, +0.547]	0.590 [0.498, 0.712]
bug. LoRA $K=32$	−0.320 [−0.507, −0.133]	1.153 [1.012, 1.523]

Table 5: World-clustered C9 key cells (binding budgets, additive mode; dense = maze-dense, rl. = rooms-large, bug. = bugtrap). The $K=16$ full-FT/LoRA CIs are disjoint (the crossover), the rooms-large $K=1$ catastrophe excludes parity, and the bugtrap LoRA degradation is significant in both metrics.